Answer Key & Teacher Notes

Session 1 — Dividing by a Unit Fraction · Session 2 — Dividing a Fraction by a Whole Number

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Dividing by a Unit Fraction

TALK PROMPTS

- How can dividing by 1/4 give an answer greater than the number you started with?
- The yarn is only 15 meters long, but 15 ÷ 1/5 = 75. Why does that make sense?

PARTNER TASK

Tape and Yarn. One partner writes 3 ÷ 1/10 for 3 yards of tape cut into 1/10-yard pieces; the other rewrites it as multiplication and solves (30). Switch roles with Craft Table: 10 meters of yarn, 1/5 meter per bracelet (50).

WATCH FOR

- Flipping the dividend instead of the divisor (4 ÷ 1/3 written as 1/4 × 1/3 = 1/12).
- Dividing by the denominator (8 ÷ 1/2 = 4): finding half of 8 instead of how many halves are in 8.
- Expecting every quotient to be smaller than the dividend.

SESSION 2 · TEACHER NOTES

Dividing a Fraction by a Whole Number

TALK PROMPTS

- Why does sharing a fraction among more people make every share smaller?
- The trail mix was already less than one pound. Why is each hiker's share still smaller than 3/4 pound?

PARTNER TASK

Lab Day. One partner finds 7/8 ÷ 5 (7/40 liter per beaker); the other checks by multiplying the share by 5. Switch roles with Fabric Strip: 6/7 meter cut into 3 pieces (2/7 meter).

WATCH FOR

- Multiplying by the whole number instead of its reciprocal (3/4 ÷ 10 written as 3/4 × 10 = 30/4).
- Flipping the fraction instead of the whole-number divisor (3/4 ÷ 3 written as 4/3 × 3).
- Sharing the whole amount before subtracting what was poured out.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S1 · Form B	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form A	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1
S2 · Form B	6.NS.A.1	1	2	1	1	2	2	3	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

unit fraction: A fraction with **1** as its numerator.

reciprocal: The number you multiply by to get a product of **1**.

quotient: The **answer** to a division problem.

- Dividing by a unit fraction asks how many **pieces** of that size fit into the number.
- Dividing by 1/4 is the same as multiplying by **4**, so 3 ÷ 1/4 = 3 × 4 = 12.
- The quotient is **greater** than the whole number you started with.

Example: Each pound holds **4** fourths. 3 pounds hold 3 × 4 = **12** fourths. Check: 12 × 1/4 = **3** pounds. Answer: 3 ÷ 1/4 = **12** balls of clay

YOU TRY

1. a. 12 b. 25

2. 2 ÷ 1/8 = 2 × 8 = 16 pieces

EXIT TICKET

1. a. 24 b. 18

2. 5 ÷ 1/3 = 5 × 3 = 15 bracelets

3. **A** (7 × 5)

PRACTICE FORM A

1. a. 16 b. 12 c. 30 d. 15 e. 40 f. 42

2. a. 8 ÷ 1/4 = 8 × 4 = 32 projects b. 2 ÷ 1/8 = 2 × 8 = 16 cubes

3. 12 ÷ 1/3 = 12 × 3 = 36 beds

4. 6 ÷ 1/8 = 6 × 8 = 48 cups

5. No. 8 ÷ 1/2 = 8 × 2 = 16 bottles. Mia found half of 8 (8 ÷ 2) instead of how many halves are in 8.

6. a. 2 ÷ 1/12 = 24 pieces b. 24 − 8 = 16 pieces

7. a. 15 ÷ 1/5 = 75 bracelets b. 15 ÷ 1/3 = 45 bracelets c. Each bracelet uses a bigger piece, so fewer pieces fit in the same 15 meters.

8. a. 9 × 3 = 27 b. Each whole holds 3 thirds, so 9 wholes hold 9 × 3 = 27 thirds. We count small pieces, so 27 > 9.

9. Rosa flipped the dividend 4 instead of the divisor 1/3. Keep 4, change ÷ to ×, flip 1/3 to 3: 4 × 3 = 12. There are 12 thirds in 4 wholes.

10. **B** (20)

11. **C** (greater than 6)

Self-check word **DIVIDE**

PRACTICE FORM B

1. a. 14 b. 8 c. 24 d. 20 e. 60 f. 40

2. a. 6 ÷ 1/3 = 6 × 3 = 18 feeders b. 3 ÷ 1/4 = 3 × 4 = 12 packets

3. 10 ÷ 1/4 = 10 × 4 = 40 beds

4. 4 ÷ 1/6 = 4 × 6 = 24 cups

5. No. 9 ÷ 1/2 = 9 × 2 = 18 bottles. Theo found half of 9 (9 ÷ 2) instead of how many halves are in 9.

6. a. 3 ÷ 1/10 = 30 pieces b. 30 − 12 = 18 pieces

7. a. 10 ÷ 1/5 = 50 bracelets b. 10 ÷ 1/2 = 20 bracelets c. Each bracelet uses a bigger piece, so fewer pieces fit in the same 10 meters.

8. a. 7 × 5 = 35 b. Each whole holds 5 fifths, so 7 wholes hold 7 × 5 = 35 fifths. We count small pieces, so 35 > 7.

9. Amir flipped the dividend 4 instead of the divisor 1/5. Keep 4, change ÷ to ×, flip 1/5 to 5: 4 × 5 = 20. There are 20 fifths in 4 wholes.

10. **D** (18)

11. **B** (9)

Self-check word **GROUPS**

GUIDED NOTES

reciprocal: Two numbers are reciprocals when their product is **1**.

multiplicative inverse: Another name for the **reciprocal**.

divisor: The number you **divide** by.

- Dividing a fraction by a whole number **shares** it into equal parts.
- To divide by 4, multiply by its reciprocal, **1/4**.
- Each share is **less** than the fraction you started with.

Example: Write 4 as a fraction: 2/5 ÷ **4/1**. Multiply by the reciprocal: 2/5 × **1/4** = 2/20. Simplify: 2/20 = **1/10**. Answer: Each person gets **1/10** pound of rice.

YOU TRY

1. a. 1/12 b. 2/7

2. 5/8 ÷ 5 = 5/8 × 1/5 = 5/40 = 1/8 of the pizza

EXIT TICKET

1. a. 1/10 b. 3/7

2. 3/4 ÷ 6 = 3/4 × 1/6 = 3/24 = 1/8 L

3. **B** (2/9 × 1/4)

PRACTICE FORM A

1. a. 1/6 b. 1/5 c. 2/5 d. 1/10 e. 4/9 f. 1/12

2. a. 3/4 ÷ 4 = 3/4 × 1/4 = 3/16 pound b. 2/3 ÷ 10 = 2/3 × 1/10 = 2/30 = 1/15 gallon

3. 3/5 ÷ 10 = 3/5 × 1/10 = 3/50 kg

4. 6/7 ÷ 3 = 6/7 × 1/3 = 6/21 = 2/7 L

5. No. 3/4 ÷ 4 = 3/4 × 1/4 = 3/16 yard. Priya multiplied by 4. Each piece must be shorter than the whole 3/4-yard string.

6. a. 7/12 − 3/12 = 4/12 = 1/3 L b. 1/3 ÷ 4 = 1/12 L

7. a. 7/8 ÷ 7 = 1/8 kg b. 7/8 ÷ 14 = 1/16 kg c. Doubling the number of trays halves each share, because the same dough is split twice as many ways.

8. 8/9 ÷ 4 = 8/9 × 1/4 = 8/36 = 2/9 m. Check: 2/9 × 4 = 8/9.

9. Caleb multiplied by 10 instead of by its reciprocal 1/10. 3/4 ÷ 10 = 3/4 × 1/10 = 3/40. The share must be less than 3/4, and 30/4 is greater.

10. **B** (1/8)

11. **A** (1/8)

Self-check word **SHARING**

PRACTICE FORM B

1. a. 1/8 b. 1/6 c. 1/4 d. 2/9 e. 1/7 f. 1/18

2. a. 5/6 ÷ 5 = 5/6 × 1/5 = 5/30 = 1/6 pound b. 3/4 ÷ 8 = 3/4 × 1/8 = 3/32 gallon

3. 2/5 ÷ 8 = 2/5 × 1/8 = 2/40 = 1/20 kg

4. 9/10 ÷ 3 = 9/10 × 1/3 = 9/30 = 3/10 L

5. No. 2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6 yard. Malik multiplied by 4. Each piece must be shorter than the whole 2/3-yard string.

6. a. 9/10 − 3/10 = 6/10 = 3/5 L b. 3/5 ÷ 3 = 1/5 L

7. a. 7/8 ÷ 5 = 7/40 L b. 7/8 ÷ 10 = 7/80 L c. Doubling the number of beakers halves each share, because the same solution is split twice as many ways.

8. 6/7 ÷ 3 = 6/7 × 1/3 = 6/21 = 2/7 m. Check: 2/7 × 3 = 6/7.

9. Jordan multiplied by 2 instead of by its reciprocal 1/2. 4/9 ÷ 2 = 4/9 × 1/2 = 4/18 = 2/9. The share must be less than 4/9, and 8/9 is greater.

10. **B** (1/5)

11. **C** (smaller than the fraction)

Self-check word **PORTION**

SHEET 1 · DIVIDING BY A UNIT FRACTION

1. 12 (each whole has 3 thirds; 4 × 3 = 12)
2. a. 12 b. 15 c. 32 d. 18 e. 21 f. 60 g. 35 h. 36
3. 6 ÷ 1/4 = 6 × 4 = 24 bags
4. 5 ÷ 1/6 = 30 pieces; 30 − 18 = 12 pieces left
5. She found half of 8 (8 ÷ 2), not how many halves are in 8. Each whole has 2 halves, so 8 ÷ 1/2 = 8 × 2 = 16.

SHEET 2 · DIVIDING A FRACTION BY A WHOLE NUMBER

1. a. 1/6 b. 2/5 c. 1/8 d. 1/9 e. 3/7 f. 1/12
2. a. 1/2 ÷ 4 = 1/8 lb b. 5/6 ÷ 5 = 1/6 L
3. 3/5 ÷ 6 = 3/5 × 1/6 = 3/30 = 1/10 gallon
4. 2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6 pound
5. less than 5/8 (5/8 ÷ 4 = 5/32; sharing into 4 parts makes each part smaller)
6. **B** (5/6 × 1/5)
7. **A** (1/9)

October 2026 conceptual additions — teacher key

Session 1 · Dividing by a Unit Fraction

Each meter holds three thirds, so 4 × 3 = 12 strips. Check 12 × 1/3 = 4 m.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Each meter holds three thirds, so 4 × 3 = 12 strips. Check 12 × 1/3 = 4 m.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Dividing a Fraction by a Whole Number

Each share is 1/4 m; 3/4 ÷ 3 = 3/4 × 1/3 = 1/4. Three shares recreate 3/4 m.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Each share is 1/4 m; 3/4 ÷ 3 = 3/4 × 1/3 = 1/4. Three shares recreate 3/4 m.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.